

Total No. of Questions : 10]

SEAT No. :

P3742

[5561]-139

[Total No. of Pages : 7

B.E. (Mechanical)

OPERATIONS RESEARCH

(2012 Course) (Semester-I) (Elective-II) (402045 C)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Figures to the right indicate full marks.
- 3) Use of scientific calculator is allowed.
- 4) Assume data wherever necessary and mention it.

Q1) Using Big M method solve,

[10]

Minimize $Z = X + Y$

Subjected to

$$2X + Y \geq 4$$

$$X + 7Y \geq 7$$

$$X, Y \geq 0$$

OR

Q2) a) Explain the assumptions underlying game theory.

[4]

- b) Four new machines M1, M2, M3, M4 are to be installed in a machine shop. There are five vacant places A, B, C, D and E available. Because of limited space, Machine M2 cannot be placed at C and M3 cannot be placed at A. Cij, the assignment cost of machine I to place j in rupees in shown below. Find the optimal assignment schedule.

[6]

	A	B	C	D	E
M1	4	6	10	5	6
M2	7	4	--	5	4
M3	--	6	9	6	2
M4	9	3	7	2	3

P.T.O.

Q3) a) What are different environments in which decisions are made? [4]

b) A plant is manufacturing 3,000 heavy duty lathes per year and is operating at 75% of its capacity. The annual sales return is Rs. 1,05,00,000. The fixed cost of the plant is Rs. 40,00,000 and variable cost is Rs. 4,150 per unit. There is a proposal to utilize spare capacity by manufacturing precision lathes which would increase the fixed cost by Rs. 8,00,000 but reduce the variable cost by Rs. 750 per unit. [6]

i) Is the proposal commercial? Give reasons supporting your answer.

ii) If the reduction in selling price by Rs. 500 per unit requires the plant to be operated at 90% of its capacity to break-even, would this be a better proposal than the earlier one?

OR

Q4) a) What is degeneracy in transportation problem? How it is resolved? [4]

b) Find the optimal strategy and value of game using game theory. [6]

	B1	B2	B3	B4
A1	1	2	-1	2
A2	3	1	2	3
A3	-1	3	2	1
A4	-2	2	0	-3

- Q5) a)** A company is considering the feasibility of changing the supplier for coupling hardware. Presently the company has an optimal policy with Ace hardware at a discount of 1%. Current yearly purchases are Rs. 81,000 and the Administrative charges are Rs. 125 per purchase and the carrying charges are 25% of the average inventory level. [8]

Bids received from other supplier are: Nutz co. offers 5% discounts if ordered twice a year and Grabbers Co. offers 3% discount if ordered four times a year. Should the company retain the present supplier or accept the proposed offers and if so, which offer?

- b)** The maintenance cost and resale value of a machine whose purchase price is Rs. 7000 is given below. When the machine should be replaced? [8]

Year	1	2	3	4	5	6	7	8
Maintenance cost in Rs.	900	1200	1600	2100	2800	3700	4700	5900
Resale value in Rs.	4000	2000	1200	600	500	400	400	400

OR

- Q6) a)** A company has a machine whose cost is Rs. 30,000. Its maintenance cost and resale value at the end of different years are given below: [10]

Year	1	2	3	4	5	6
Maintenance cost in Rs.	4500	4700	5000	5500	6500	7500
Resale value in Rs.	27000	25300	24000	21000	18000	13000

- i) What is the economic life of the machine and what is the minimum average cost?
- ii) The company has obtained a contract to supply goods produced by the machine for 5 years from now. After 5 years, the company does not intend to use the machine. If the machine at present is one year old, what replacement policy should the company adopt if it intends to replace the machine not more than once?
- b) A newspaper boy buys paper for 5 rupees each and sells them for 6 rupees each. He cannot return unsold newspapers. Daily demand 'R' for the newspaper follows the distribution given in table. If each day's demand is independent of the previous days, how many papers should be ordered each day? [6]

Demand (R)	10	11	12	13	14	15	16
Probability (P_R)	0.05	0.15	0.40	0.20	0.10	0.05	0.05

- Q7) a)** Arrival rate of a telephone calls at a telephone booth are according to Poisson distribution with an average time of 9 minutes between two consecutive arrivals. The length of telephone call is assumed to be exponentially distributed with mean 3 minutes. [10]
- i) Determine the probability that a person arriving at the booth will have to wait.
- ii) Find the average queue length that is formed from time to time.

- iii) The telephone company will install a second booth when convinced that an arrival would expect to have wait at least four minutes for the phone. Find the increase in flow rate of arrivals which will justify the second booth.
- iv) What is the probability that an arrival will have to wait for more than 10 minutes before the phone is free?
- v) Find the fraction of the day that the phone will be in use.
- b) There are seven jobs, each of which has to go through the machine A and B in the order AB. Processing time in hours is given below. Determine the sequence of these jobs that will minimize the total elapsed time T. Also find T and idle time for the machines A and B. [6]

Jobs	1	2	3	4	5	6	7
Machine A	3	12	15	6	10	11	9
Machine B	8	10	10	6	12	1	3

OR

- Q8)** a) A repairman is to be hired to repair machines which break down at an average rate of 3 per hour. The breakdown follows Poisson distribution. Non-productive time of machine is considered to cost Rs. 16 per hour. Two repairman have been interviewed: one is slow but cheap while the other is fast but expensive. The slow repairman charges Rs. 8 per hour and he services breakdown machines at 4 per hour. The fast repairman demands Rs. 20 per hour and he services at an average rate of 6 per hour. Which repairman should be hired? Assume an 8 hour working day. [7]
- b) There are five jobs, each of which is to be processed through three machines A, B, C in the order ABC. The processing time in hours is as below. Determine the optimum sequence for the five jobs and the minimum elapsed. Also find the idle time for the three machines and waiting time for the jobs. [9]

Jobs	1	2	3	4	5
Machine A	3	8	7	5	4
Machine B	4	5	1	2	3
Machine C	7	9	5	6	10

Q9) a) A project schedule has the following characteristics:

[8]

Activity	Time (Weeks)	Activity	Time (Weeks)
1-2	4	5-6	4
1-3	1	5-7	8
2-4	1	6-8	1
3-4	1	7-8	2
3-5	6	8-10	5
4-9	5	9-10	7

- Construct the project network.
 - Compute the E and L for each event.
 - Find the critical path and project duration.
- b) In a workshop, fitters report at the tool crib after 8 am to get the tools where room attendant is present to issue them. The arrival and service time distribution are shown below. **[10]**

Arrival Distribution		Service Distribution	
Inter-arrival time (Min)	Frequency	Service Time (Min)	Frequency
3.5-4.5	0.05	3.5-4.5	0.10
4.5-5.5	0.20	4.5-5.5	0.20
5.5-6.5	0.35	5.5-6.5	0.40
6.5-7.5	0.25	6.5-7.5	0.20
7.5-8.5	0.10	7.5-8.5	0.10
8.5-9.5	0.05		

Determine the average waiting time of a fitter in the queue. What is the average time he spends in the system. Simulate the queuing phenomenon for the period of 2 hours. Also find the average waiting time in non-empty queue and percent time the tool room attendant remains busy in this period. Use following random numbers.

Random Numbers for Arrival time: 15, 09, 41, 74, 72, 67, 55, 71, 35, 41, 96, 20, 45, 38, 01, 67, 63, 39, 55

Random numbers for service time: 20, 72, 34, 54, 40, 31, 22, 48, 74, 76, 02, 07, 64, 95, 23, 91, 48, 55, 91

OR

Q10)a) Explain the application of simulation techniques in inventory problem.[4]

b) An engineering firm has to bid for a project. The activities and their time estimates are as follows: [14]

Activity	Optimistic	Most Likely	Pessimistic
1-2	14	17	25
2-3	14	18	21
2-4	13	15	18
2-8	16	19	28
3-4	0	0	0
3-5	15	18	27
4-6	13	17	21
5-7	0	0	0
5-9	14	18	20
6-7	0	0	0
6-8	0	0	0
7-9	16	20	41
8-9	14	16	22

The policy of the firm with respect to submitting bids is to bid the minimum amount that will provide a 95% of probability of the best breaking even. The fixed cost of the project are Rs. 8,00,000 and the variable cost are Rs. 9,000 per day spent working on the project. The duration is given in days and cost is in rupees. What amount firm should bid under this policy?

